FAHEEMA TAMTON

AI / Machine Learning Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with experience designing, training, testing, and deploying intelligent systems using Machine Learning, Deep Learning, Computer Vision, NLP, and Generative AI. Skilled in building scalable AI solutions with LLMs, VLMs, OCR, RAG, TensorFlow, PyTorch, and Python for real-world applications and automation.

TECHNICAL SKILLS

- **Programming & Development:** Python, SQL, Java, JavaScript, HTML, CSS, C, Git/GitHub, VS Code, PyCharm, Jupyter Notebook, Google Colab, Anaconda
- **Machine Learning & AI:** Supervised Unsupervised Learning, Neural Networks (CNNs, RNNs, Transformers), Feature Engineering, Model Fine-Tuning, Prompt Engineering, LLM Integration (Gemini, OpenAI API)
- **Computer Vision NLP:** OpenCV, OCR Pipelines, Object Detection, Image Processing, Natural Language Processing, NLTK, spaCy
- **Frameworks & Libraries:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Flask, FastAPI, Matplotlib, Seaborn
- **Data Engineering & Analytics:** Data Preprocessing, ETL Pipelines, Data Visualization, RESTful API Design, Structured Data Extraction
- **Deployment & Tools:** Roboflow, Kaggle, Git/GitHub, Google Colab, Inference Optimization, Docker

AI / ML PROJECTS

Easy Heal — Full-Stack AI Web Application — *AI, Qwen 2.5 VLM, NLP, DL, CV*

- Developed an AI-driven prescription intelligence system using Vision-Language Models (VLMs), OCR, multimodal inference, and NLP-based entity extraction to interpret handwritten prescriptions and generate structured medical data with 85% extraction accuracy.
- Engineered a scalable healthcare automation backend using Django ORM, Python scheduling algorithms, cron-based real-time alerting, and adherence tracking, enabling intelligent multi-patient medication scheduling and real-time healthcare workflow automation.

Medicine Blister Text Extraction — Apollo Pharmacy — *VLMs, PaddleOCR, EasyOCR, Roboflow*

- Developed an AI-driven document intelligence pipeline for extracting and validating medicine blister text with automated expiry date recognition using OCR and Vision-Language Models.
- Conducted VLM vs OCR performance evaluation and handled dataset annotation, preprocessing, and optimisation workflows using Roboflow.

Object Detection for Semi-Humanoid Robots — I HUB Research & Robotics Pvt. Ltd. — *Roboflow*

- Handled and preprocessed large-scale image datasets, including annotation and data augmentation, and implemented YOLO-based object detection to enable accurate multi-object recognition while optimizing data pipelines, training workflows, and inference performance for semi-humanoid robotic vision systems.
- Achieved 80% mAP with inference time of 4[illegible]0 ms per image.

InMyPDF — Multimodal AI — *LLM, RAG, FAISS, OpenAI*

- Built a Retrieval-Augmented Generation (RAG) system to answer user queries from PDF documents using an LLM grounded in retrieved context, implemented PDF ingestion, recursive text chunking, embedding generation, and FAISS-powered retrieval. Integrated top-k context retrieval with an OpenAI language model to generate accurate, context-aware responses while minimizing hallucinations.
- Achieved 85% top-k retrieval precision, improving response accuracy and reducing hallucinations.

Data Pipeline for AI/ML Applications — I HUB Research & Robotics Pvt. Ltd.

- Designed and implemented end-to-end data pipelines to collect, preprocess, and manage real-time multi-source datasets using web scraping, streaming ingestion, and data quality validation to prepare structured data for model training, evaluation, and deployment in AI/ML applications.

SoulMirror — *ML, Python, Flask*

- Built an ML-based system to predict MBTI personality types from questionnaire responses. Implemented preprocessing, feature encoding, classification, and inference pipelines. Integrated the model into a Flask application to deliver personalized personality insights.
- Achieved 78% classification accuracy using optimized preprocessing and feature engineering.

PROFESSIONAL EXPERIENCE

AI / ML Engineer Intern | **I HUB Research & Robotics Private Limited** *Sep 2024 – Mar 2026*

- Worked in a team of 3 members on AI/ML and computer vision projects for real-world applications.
- Handled large-scale datasets (25000+ images), including annotation, preprocessing, and augmentation.
- Collaborated on production-level systems involving OCR, object detection, and data pipelines.
- Developed real-time computer vision solutions for semi-humanoid robotics using YOLO; including dataset preparation, model training, performance evaluation, and optimisation for deployment-ready inference.

Python Intern | **Kiwisoft Solutions LLP** *Oct 2024 – Mar 2025*

- Developed full-stack web applications using Python, designing RESTful APIs, implementing authentication, managing databases, and integrating UPI-based payment systems.

EDUCATION

Post Graduate Diploma in Artificial Intelligence & Machine Learning — *I Hub School of Learning*

Bachelor of Science in Computer Science — *Kannur University, Kerala*

CERTIFICATIONS

- AI For Everyone by Andrew Ng — DeepLearning.ai (Coursera) *View Verified Credential*
- AI Fundamentals: Foundations for Understanding AI — IBM SkillsBuild *View Verified Credential*
- Introduction to Generative AI — Google Cloud (Skill Badge) *View Verified Credential*
- Machine Learning using Python — Simplilearn SkillUp *View Verified Credential*